



ME 342A: SYSTEM DESIGN AND ANALYSIS

UNIT 4 SYSTEM SCHEDULE

Learning outcomes

- **Time Models**
- **Resource Allocation**
- **Time Cost Trade Off**
- **System Cost Economic Flow Graph**



1 Make or Buy Decision

Depends on

- Volume
- Fixed cost of making
- Per-unit direct cost when making
- Per-unit cost when buying

Reason for Making

- Cost concerns
- Desire to expand the manufacturing focus
- Need of direct control over the product
- Intellectual property concerns
- Quality control concerns
- Supplier unreliability
- Lack of competent suppliers
- Volume too small to get a supplier attracted
- Reduction of logistic costs (shipping etc.)
- To maintain a backup source
- Organizational pride

Reason for Buying

- Lack of technical experience
- Supplier's expertise on the technical areas and the domain
- Cost considerations
- Need of small volume
- Insufficient capacity to produce in-house
- Brand preferences
- Strategic partnerships



Approaches for Make or Buy Decisions

- a) Simple Cost Analysis
- b) Economic Analysis
- c) Break Even Analysis

a) Simple Cost Analysis

Simple Cost Analysis is a mathematical approach to compare the costs and expected benefits of two or more projects (or options). It is a decision-making tool widely used in economics and applicable to many industries such as manufacturing, IT, software development, construction, education, healthcare, and information technology. The main purpose of tracking the Simple Cost Analysis steps is to calculate the ratio of benefits over costs. It is a simple technique that can be used for financial decisions. It is conducted to identify how well, or how poorly, a project will be concluded.

b) Economic Analysis

Based on two inventory models

- 1. Purchase Model for Economic Order Quantity (EOQ or Q System)
- 2. Manufacturing Model for Economic Production Quantity (EPQ)

EOQ Equations

$$TC_{EOQ} = \left(\frac{D}{Q}S\right) + \left(\frac{Q}{2}H\right)$$

Where

TC = total annual cost

D = annual demand

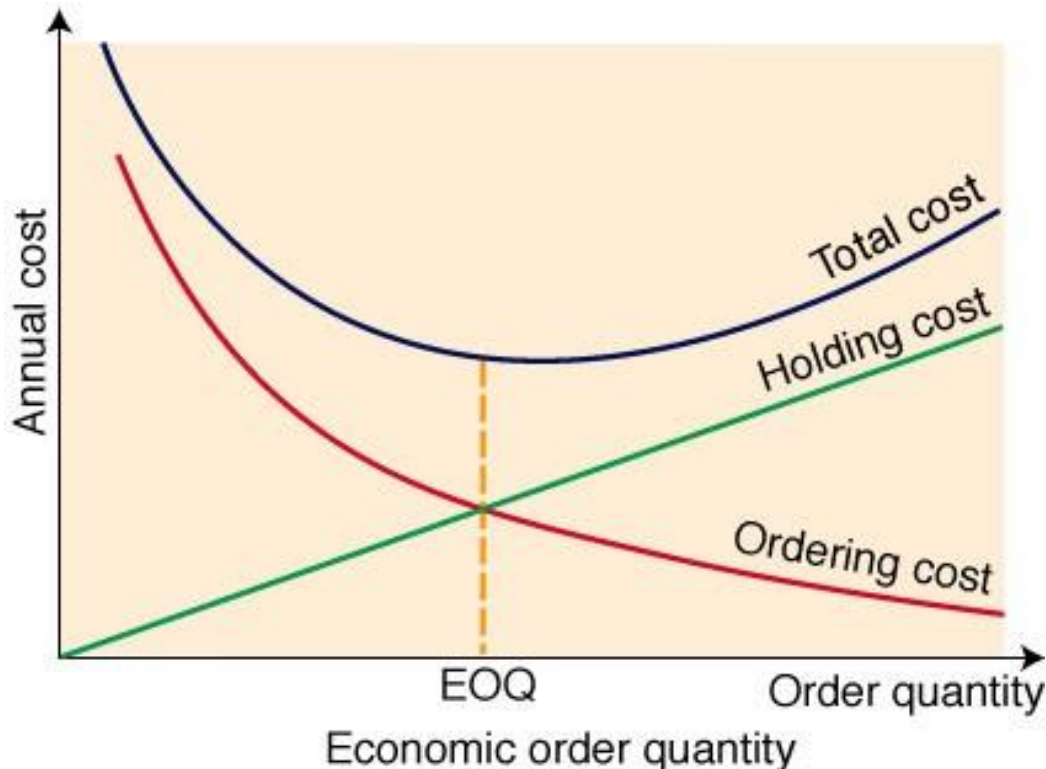
Q = quantity to be ordered

H = annual holding cost

S = ordering or setup cost

EOQ Total Costs

Total annual costs = annual ordering costs + annual holding costs



The EOQ Formula

Minimize the TC by ordering the EOQ:

$$EOQ = \sqrt{\frac{2DS}{H}}$$

When to Order:

The Reorder Point

- Without safety stock:

$$R = dL$$

where R = reorder point in units

d = daily/weekly demand in units

L = lead time in days/weeks

- With safety stock:

$$R = dL + SS$$

where SS = safety stock in units

EOQ Example

Weekly demand = 240 units



No. of weeks per year = 52

Ordering cost = Rs. 50

Unit cost = Rs. 15

Annual carrying charge = 20%

Lead time = 2 weeks

Solution:

$$D = 52 \times 240 = 12,480 \text{ units/year}$$

$$H = 0.2 \times 15 = \text{Rs. } 3 \text{ per unit per year}$$

$$Q = \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2 \times 12,480 \times 50}{3}} = 644.98 \cong 645 \text{ units}$$

$$\begin{aligned} TC &= \left(\frac{D}{Q} S \right) + \left(\frac{Q}{2} H \right) = \left(\frac{12,480}{645} \times 50 \right) + \left(\frac{645}{2} \times 3 \right) \\ &= 967.44 + 967.5 = \text{Rs. } 1,934.94 \end{aligned}$$

$$R = dL = 240 \times 2 = 480 \text{ units}$$

EPQ Equations

- Adjusted total cost: $TC_{EPQ} = \left(\frac{D}{Q} S \right) + \left(\frac{I_{MAX}}{2} H \right)$
- Maximum inventory: $I_{MAX} = Q \left(1 - \frac{d}{p} \right)$
- Adjusted order quantity: $EPQ = \sqrt{\frac{2DS}{H \left(1 - \frac{d}{p} \right)}}$

EPQ Example

Annual demand = 18,000 units

Production rate = 2500 units/month

Setup cost = Rs. 800

Annual holding cost = Rs. 18 per unit

Lead time = 5 days



No. of operating days per month = 20

Solution:

$$d = \frac{18,000}{12} = 1500 \text{ Units/month; } p = 2500 \text{ Units/month}$$

$$Q = \sqrt{\frac{2DS}{H\left(1 - \frac{d}{p}\right)}} = \sqrt{\frac{2 \times 18,000 \times 800}{18 \times \left(1 - \frac{1500}{2500}\right)}} = 2000 \text{ units}$$

$$I_{\text{MAX}} = Q\left(1 - \frac{d}{p}\right) = 2000 \times \left(1 - \frac{1500}{2500}\right) = 800 \text{ units}$$

$$TC = \left(\frac{D}{Q}S\right) + \left(\frac{I_{\text{MAX}}}{2}H\right) = \left(\frac{18,000}{2000} \times 800\right) + \left(\frac{800}{2} \times 18\right)$$

$$= 7,200 + 7,200 = 14,400$$

The reorder point:

$$R = dL = \frac{1500}{20} \times 5 = 375 \text{ units}$$

With safety stock of 200 units:

$$R = dL + SS = \frac{1500}{20} \times 5 + 200 = 575 \text{ units}$$

c) Break-Even Analysis: Graphical Approach

Compute quantity of goods that must be sold to break-even

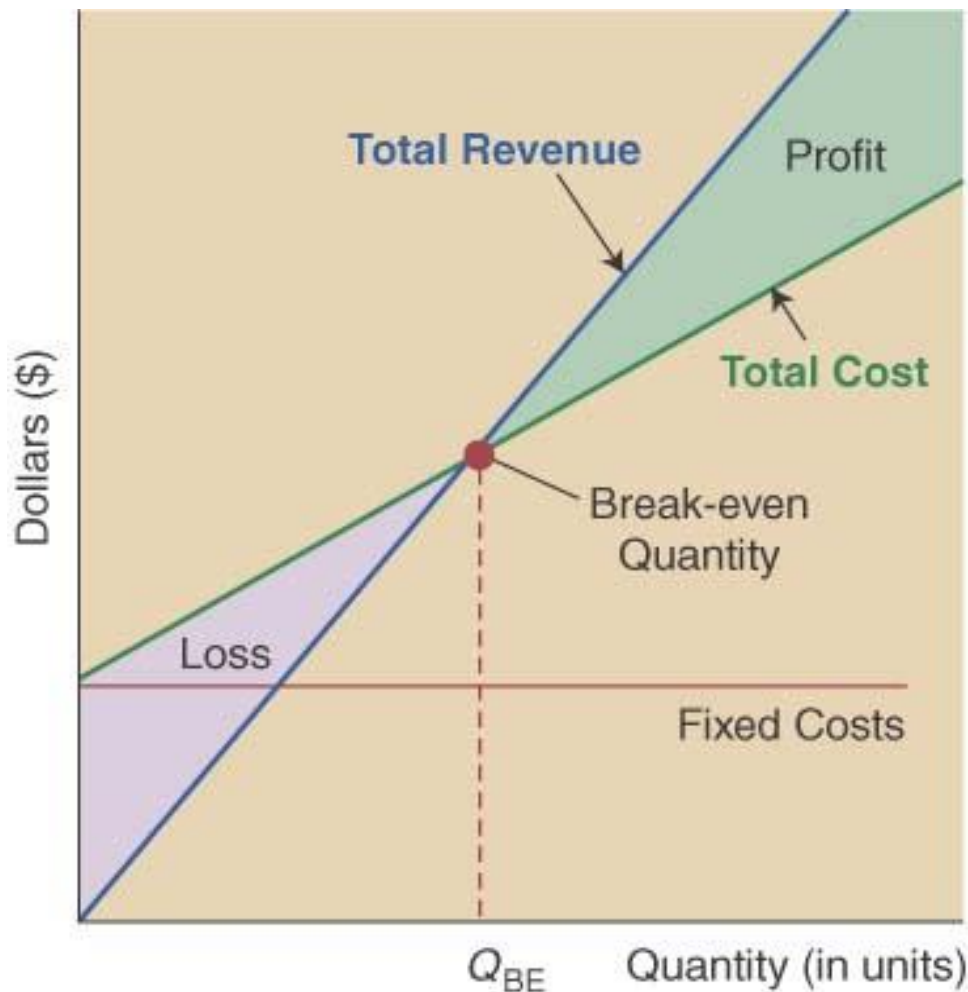
Compute total revenue at an assumed selling price

Compute fixed cost and variable cost for several quantities

Plot the total revenue line and the total cost line

Intersection is break-even

Sensitivity analysis can be done to examine changes



Total cost = fixed costs + variable costs (quantity):

$$TC = F + (VC)Q$$

Revenue = selling price (quantity)

$$R = (SP)Q$$

Break-even point is where total costs = revenue:

$$TC = R \quad \text{or} \quad F + (VC)Q = (SP)Q$$

$$\text{or} \quad Q = \frac{F}{SP - VC}$$

2 Types of Forecasting Models

1. Qualitative methods – judgmental methods
 - a. Forecasts generated subjectively by the forecaster
 - b. Educated guesses



2. Quantitative methods – based on mathematical modeling:

a. Forecasts generated through mathematical modeling

2.1 Qualitative methods

Type	Characteristics	Strengths	Weaknesses
Executive opinion	A group of managers meet & come up with a forecast	Good for strategic or new-product forecasting	One person's opinion can dominate the forecast
Market research	Uses surveys & interviews to identify customer preferences	Good determinant of customer preferences	It can be difficult to develop a good questionnaire
Delphi method	Seeks to develop a consensus among a group of experts	Excellent for forecasting long-term product demand, technological	Time consuming to develop

2.2 Quantitative methods

A. Time Series Models:

- Assumes information needed to generate a forecast is contained in a time series of data
- Assumes the future will follow same patterns as the past

B. Causal Models or Associative Models

- Explores cause-and-effect relationships
- Uses leading indicators to predict the future

2.2.1 Time series models

Forecaster looks for data patterns as

$\text{Data} = \text{historic pattern} + \text{random variation}$

Historic pattern to be forecasted:

Level (long-term average) – data fluctuates around a constant mean

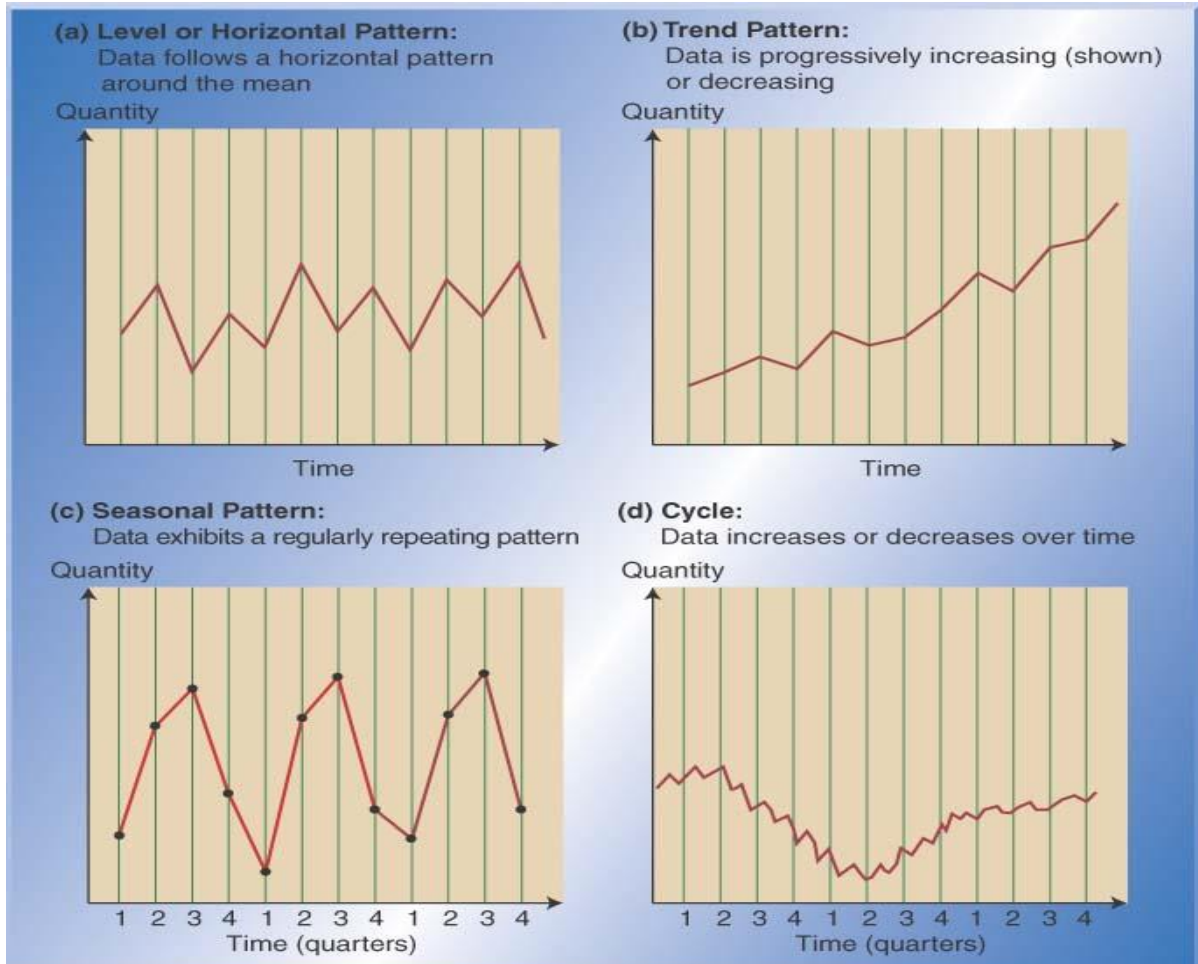
Trend – data exhibits an increasing or decreasing pattern

Seasonality – any pattern that regularly repeats itself and is of a constant length

Cycle – patterns created by economic fluctuations

Random Variation cannot be predicted

Time Series Patterns



Time Series Models

- Naive: $F_{t+1} = A_t$
 - The forecast is equal to the actual value observed during the last period – good for level patterns
- Simple Mean: $F_{t+1} = \sum A_t / n$
 - The average of all available data - good for level patterns
- Moving Average: $F_{t+1} = \sum A_t / n$
 - The average value over a set period (e.g.: the last four weeks)



- Each new forecast drops the oldest data point & adds a new observation
- More responsive to a trend but still lags actual data
- Weighted Moving Average: $F_{t+1} = \sum C_t A_t$

All weights must add to 100% or 1.00

e.g. **Ct .5**, **Ct-1 .3**, **Ct-2 .2** (weights add to 1.0)

Allows emphasizing one period over others; above indicates more weight on recent data (Ct=.5)

Differs from the simple moving average that weighs all periods equally - more responsive to trends

- Exponential Smoothing: $F_{t+1} = \alpha A_t + (1 - \alpha)F_t$

Most frequently used time series method because of ease of use and minimal amount of data needed

Need just three pieces of data to start:

Last period's forecast (**Ft**)

Last periods actual value (**At**)

Select value of smoothing coefficient between 0 and 1.0

If no last period forecast is available, average the last few periods or use naive method

Higher values (e.g. .7 or .8) may place too much weight on last period's random variation

Linear Trend Line

A time series technique that computes a forecast with trend by drawing a straight line through a set of data using this formula:

$$Y = a + bx$$

where

Y = forecast for period X

X = the number of time periods from X = 0



A = value of y at X = 0 (Y intercept)

B = slope of the line

Forecasting Trend

Basic forecasting models for trends compensate for the lagging that would otherwise occur.

One model, **trend-adjusted exponential smoothing** uses a three-step process

Step 1 - Smoothing the level of the series

$$S_t = \alpha A_t + (1 - \alpha)(S_{t-1} + T_{t-1})$$

Step 2 – Smoothing the trend

$$T_t = \beta(S_t - S_{t-1}) + (1 - \beta)T_{t-1}$$

Forecast including the trend

$$FIT_{t+1} = S_t + T_t$$

Forecasting Seasonality

- a) Calculate the average demand per season
 - a. *E.g.*: average quarterly demand
- b) Calculate a seasonal index for each season of each year:
 - a. Divide the actual demand of each season by the average demand per season for that year
- c) Average the indexes by season
 - a. *E.g.*: take the average of all Spring indexes, then of all Summer indexes
- d) Forecast demand for the next year and divide by the number of seasons
 - a. Use regular forecasting method and divide by four for average quarterly demand
- e) Multiply next year's average seasonal demand by each average seasonal index

Result is a forecast of demand for each season of next year

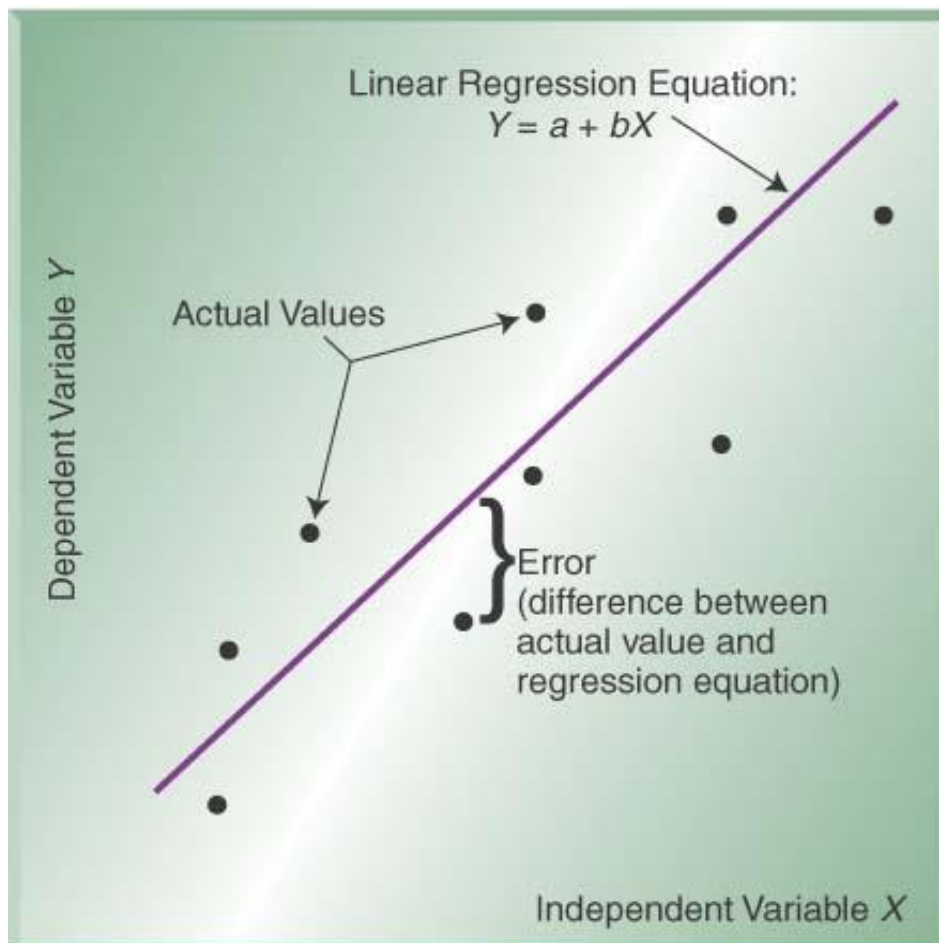
2.2.2 Causal models

Often, leading indicators can help to predict changes in future demand. Causal models establish a cause-and-effect relationship between independent and dependent variables. A common tool of causal modeling is linear regression:

$$Y = a + bx$$

Additional related variables may require multiple regression modeling

Linear Regression



Identify dependent (**y**) and independent (**x**) variables

Solve for the slope of the line

$$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - n\bar{X}^2}$$

Solve for the y intercept

$$a = \bar{Y} - b\bar{X}$$



Develop your equation for the trend line

$$Y = a + bX$$

Measuring Forecast Error

Forecasts are never perfect. We need to know how much we should rely on our chosen forecasting method. Forecast error can be estimating as:

$$E_t = A_t - F_t$$

over-forecasts = negative errors

under-forecasts = positive errors

Measuring Forecasting Accuracy

Following are the methods to measure forecasting accuracy

1. Mean Absolute Deviation (MAD)

$$MAD = \frac{\sum |actual - forecast|}{n}; \text{ measures the total error in a forecast without regard to sign}$$

2. Cumulative Forecast Error (CFE)

$$CFE = \sum (actual - forecast); \text{ measures any bias in the forecast}$$

3. Mean Square Error (MSE)

$$MSE = \frac{\sum (actual - forecast)^2}{n}; \text{ Penalizes larger errors}$$

4. Tracking Signal

$$TS = \frac{CFE}{MAD}; \text{ Measures if your model is working}$$

3 Objectives of Scheduling

3.1 Maximization of the utilization of resources

For a finite set of tasks, the utilization of the resources is inversely proportional to the time required to accomplish all the tasks. This time is known as the makespan or maximum flow time of the schedule in a static scheduling system.



In finite scheduling problems, resource utilization is improved by scheduling a set of tasks to reduce makespan.

3.2 Minimization of the waiting in process inventory

The reduction of the average number of tasks waiting in a queue while the resources is busy with other tasks. If the jobs spend less time in the system, the inventory is reduced. In scheduling terms, this leads to minimizing the average flow time (mean flow time) or the average of the times the jobs spend in the system. The makespan of a scheduling is constant, the sequence that reduces mean flow time also reduces mean in-process inventory.

3.3 Minimization of tardiness

In many situations, some or all the tasks have due dates and a penalty is incurred if a task is finished after that due date. The possible objectives relating to tardiness are the minimization of maximum tardiness, the minimization of the number of tasks that are tardy, or the minimization of mean tardiness. Sometimes, completion of the jobs ahead of the due dates is also undesirable. For such scheduling problems, the objective is to minimize a penalty function of earliness and tardiness. Here, it can minimize either the number of early/tardy jobs or minimize the maximum earliness/tardiness. If all the jobs have a common due date, this is usually achieved by minimizing the squared deviation of the job completion times about the due date. If the analyst can fix the due date, the problem becomes one of minimizing the variance of completion times.

4 Priority Rules to Select Order of Jobs Waiting for Processing

Sequencing is concerned with determining the order in which jobs are processed. Not only must the order be determined for processing jobs at work centers but also for work processed at individual workstations. When work centers are heavily loaded and lengthy jobs are involved, the situation can become complicated. The order of processing can be crucial when it comes to the cost of waiting to be processed and the cost of idle time at work centers.



There are several priority rules or heuristics that can be used to select the order of jobs waiting for processing:

- Random (R). Pick any job in the queue with equal probability. This rule is often used as a benchmark for other rules.
- First come/first served (FC/FS). This rule is sometimes deemed to be fair since jobs are processed in the order in which they arrive.
- Shortest processing time (SPT). The job with the shortest processing time requirement goes first. This rule tends to reduce work-in-process inventory, average throughput time, and average job lateness.
- Earliest due date (EDD). The job with the earliest due date goes first. This seems to work well if the firm performance is judged by job lateness.
- Critical ratio (CR). To use this rule, one must calculate a priority index using the formula $(\text{due date} - \text{now}) / (\text{lead time remaining})$. This rule is widely used in practice.
- Least work remaining (LWR). An extension of SPT, this rule dictates that work be scheduled according to the processing time remaining before the job is complete. The less work remaining in a job, the earlier it is in the production schedule.
- Fewest operations remaining (FOR). This rule is another variant of SPT; it sequences jobs based on the number of successive operations remaining until the job is considered complete. The fewer operations that remain, the earlier the job is scheduled.
- Slack time (ST). This rule is a variant of EDD; it utilizes a variable known as slack. Slack is computed by subtracting the sum of setup and processing times from the time remaining until the job's due date. Jobs are run in order of the smallest amount of slack.



- Slack time per operation (ST/O). This is a variant of ST. The slack time is divided by the number of operations remaining until the job is complete with the smallest values being scheduled first.
- Next queue (NQ). NQ is based on machine utilization. The idea is to consider queues (waiting lines) at each of the succeeding work centers at which the jobs will go. One then selects the job for processing that is going to the smallest queue, measured either in hours or jobs.
- Least setup (LSU). This rule maximizes utilization. The process calls for scheduling first the job that minimizes changeover time on a given machine.

5 Critical Path in Network Analysis

For basic scheduling computations, the notations used are

(i, j) = Activity with tail event i and head event j

E_i = Earliest occurrence time of event i

L_j = Latest allowable occurrence time of event j

D_{ij} = Estimated completion time of activity (i, j)

$(E_s)_{ij}$ = Earliest starting time of activity (i, j)

$(E_f)_{ij}$ = Earliest finishing time of activity (i, j)

$(L_s)_{ij}$ = Latest starting time of activity (i, j)

$(L_f)_{ij}$ = Latest finishing time of activity (i, j)

The procedure is as follows:

1. Determination of Earliest time (E_j): Forward Pass computation

Step 1: The computation begins from the start node and move towards the end node. For easiness, the forward pass computation starts by assuming the earliest occurrence time of zero for the initial project event.

Step 2:

- I. Earliest starting time of activity (i, j) is the earliest event time of the tail end event i.e. $(E_s)_{ij} = E_i$



- II. Earliest finish time of activity (i, j) is the earliest starting time + the activity time i.e. $(Ef)_{ij} = (Es)_{ij} + D_{ij}$ or $(Ef)_{ij} = E_i + D_{ij}$
- III. Earliest event time for event j is the maximum of the earliest finish times of all activities ending in to that event i.e. $E_j = \max [(Ef)_{ij} \text{ for all immediate predecessor of } (i, j)]$ or $E_j = \max [E_i + D_{ij}]$

2. Backward Pass computation (for latest allowable time)

Step 1: For ending event assume $E = L$. Remember that all E's have been computed by forward pass computations.

Step 2: Latest finish time for activity (i, j) is equal to the latest event time of event j i.e. $(Lf)_{ij} = L_j$

Step 3: Latest starting time of activity (i, j) = the latest completion time of (i, j) – the activity time or $(Ls)_{ij} = (Lf)_{ij} - D_{ij}$ or $(Ls)_{ij} = L_j - D_{ij}$

Step 4: Latest event time for event 'i' is the minimum of the latest start time of all activities originating from that event i.e. $L_i = \min [(Ls)_{ij} \text{ for all immediate successor of } (i, j)] = \min [(Lf)_{ij} - D_{ij}] = \min [L_j - D_{ij}]$

3. Determination of floats and slack times

There are three kinds of floats

- a. Total float: The amount of time by which the completion of an activity could be delayed beyond the earliest expected completion time without affecting the overall project duration time. Mathematically

$$(Tf)_{ij} = (\text{Latest start} - \text{Earliest start}) \text{ for activity } (i - j)$$

$$(Tf)_{ij} = (Ls)_{ij} - (Es)_{ij} \text{ or } (Tf)_{ij} = (L_j - D_{ij}) - E_i$$

- b. Free float: The time by which the completion of an activity can be delayed beyond the earliest finish time without affecting the earliest start of a subsequent activity. Mathematically

$$(Ff)_{ij} = (\text{Earliest time for event } j - \text{Earliest time for event } i) - \text{Activity time for } (i, j)$$

$$(Ff)_{ij} = (E_j - E_i) - D_{ij}$$



- c. Independent float: The amount of time by which the start of an activity can be delayed without effecting the earliest start time of any immediately following activities if the preceding activity has finished at its latest finish time. Mathematically

$$(If)_{ij} = (E_j - L_i) - D_{ij}$$

The negative independent float is always taken as zero.

Event slack: It is defined as the difference between the latest event and earliest event times. Mathematically

$$\text{Head event slack} = L_j - E_j, \text{ Tail event slack} = L_i - E_i$$

Determination of critical path

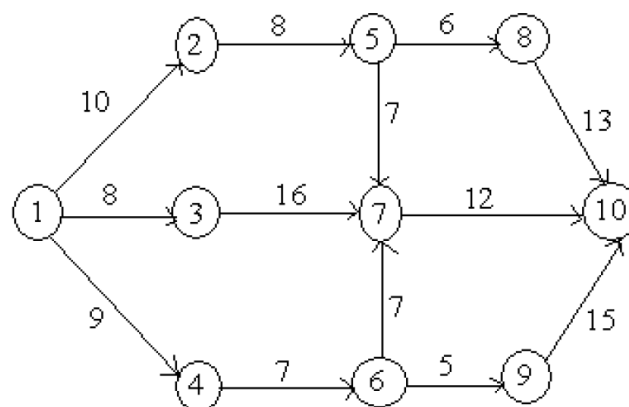
Critical event: The events with zero slack times are called critical events. In other words, the event i is said to be critical if $E_i = L_i$

Critical activity: The activities with zero total float are known as critical activities. In other words, an activity is said to be critical if a delay in its start will cause a further delay in the completion date of the entire project.

Critical path: The sequence of critical activities in a network is called critical path. The critical path is the longest path in the network from the starting event to ending event and defines the minimum time required to complete the project.

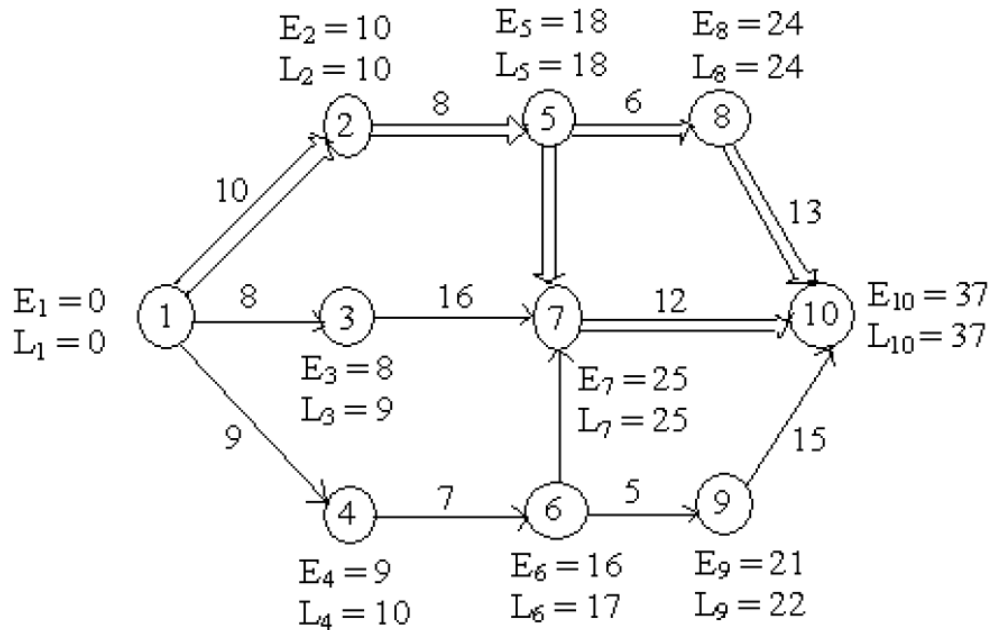
Example 1

Determine the early start and late start in respect of all node points and identify critical path for the following network



Solution:

Calculation of E and L for each node is shown in the network



Activity(i, j)	Normal Time (D_{ij})	Earliest Time		Latest Time		Float Time ($(L_i - D_{ij}) - E_i$)
		Start (E_i)	Finish ($E_i + D_{ij}$)	Start ($L_i - D_{ij}$)	Finish (L_i)	
(1, 2)	10	0	10	0	10	0
(1, 3)	8	0	8	1	9	1
(1, 4)	9	0	9	1	10	1
(2, 5)	8	10	18	10	18	0
(4, 6)	7	9	16	10	17	1
(3, 7)	16	8	24	9	25	1
(5, 7)	7	18	25	18	25	0
(6, 7)	7	16	23	18	25	2
(5, 8)	6	18	24	18	24	0
(6, 9)	5	16	21	17	22	1
(7, 10)	12	25	37	25	37	0
(8, 10)	13	24	37	24	37	0
(9, 10)	15	21	36	22	37	1

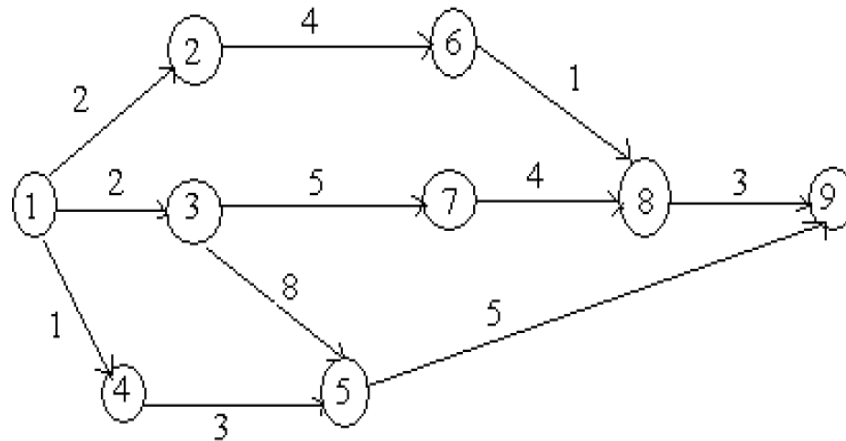
From the table, the critical nodes are (1, 2), (2, 5), (5, 7), (5, 8), (7, 10) and (8, 10). There are two possible critical paths

$$1 \rightarrow 2 \rightarrow 5 \rightarrow 8 \rightarrow 10$$

$$1 \rightarrow 2 \rightarrow 5 \rightarrow 7 \rightarrow 10$$

Example 2

Find the critical path and calculate the slack time for the following network



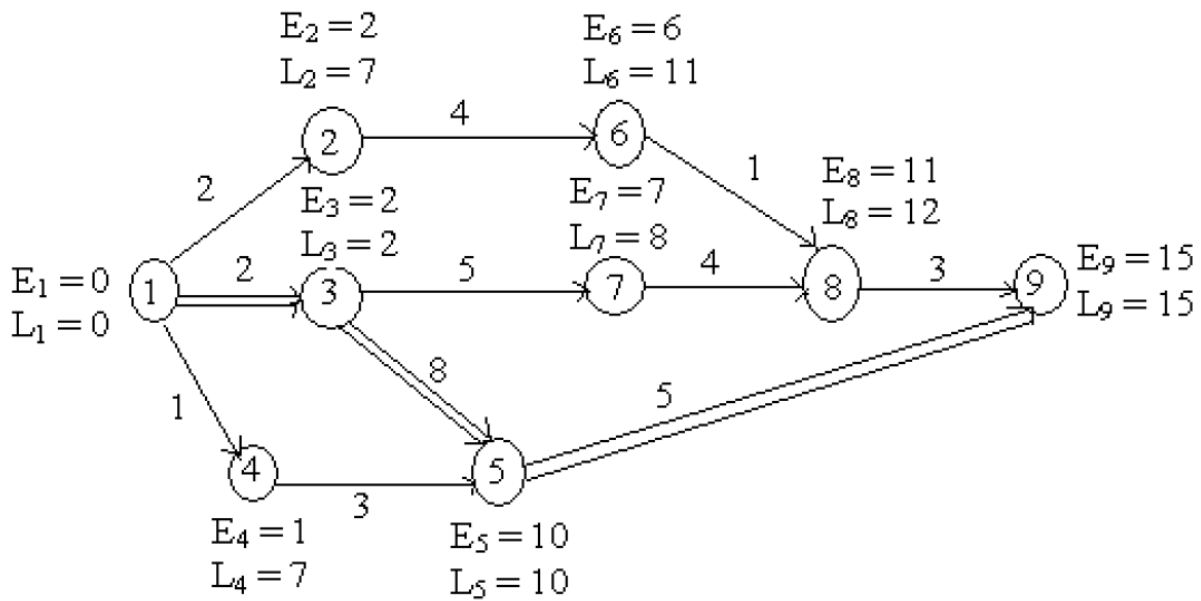
Solution

The earliest time and the latest time are obtained below.

Activity(i, j)	Normal Time (D _{ij})	Earliest Time		Latest Time		Float Time (L _i - D _{ij}) - E _i
		Start (E _i)	Finish (E _i + D _{ij})	Start (L _i - D _{ij})	Finish (L _i)	
(1, 2)	2	0	2	5	7	5
(1, 3)	2	0	2	0	2	0
(1, 4)	1	0	1	6	7	6
(2, 6)	4	2	6	7	11	5
(3, 7)	5	2	7	3	8	1
(3, 5)	8	2	10	2	10	0
(4, 5)	3	1	4	7	10	6
(5, 9)	5	10	15	10	15	0
(6, 8)	1	6	7	11	12	5
(7, 8)	4	7	11	8	12	1
(8, 9)	3	11	14	12	15	1

From the above table, the critical nodes are the activities (1, 3), (3, 5) and (5, 9)

The critical path is 1 → 3 → 5 → 9



The critical path is $1 \rightarrow 3 \rightarrow 5 \rightarrow 9$

6 Resource Allocation and Leveling

Aggregate production planning is about the allocation of resources. Assuming that a plan is basically accurate, a production plan, usually stretching no more than a year into the future, will ensure the smooth transition of production capacity as demand changes over time. The significance of this is that once employees are habituated to these changes, they will alter their own schedules and work habits to reflect changes in demand. This means that costs of changing work schedules will be minimized, increasing efficiency.

There are two types of resource problem

Resource smoothing

Resource levelling

6.1 Resource smoothing

There may be a ceiling on the availability of resources in a period. For instance, only Rs 125 lakh per annum may be available to the project and if unutilized during the year, the remaining amount lapses. The resource analysis used for this type of case is called Resource smoothing.



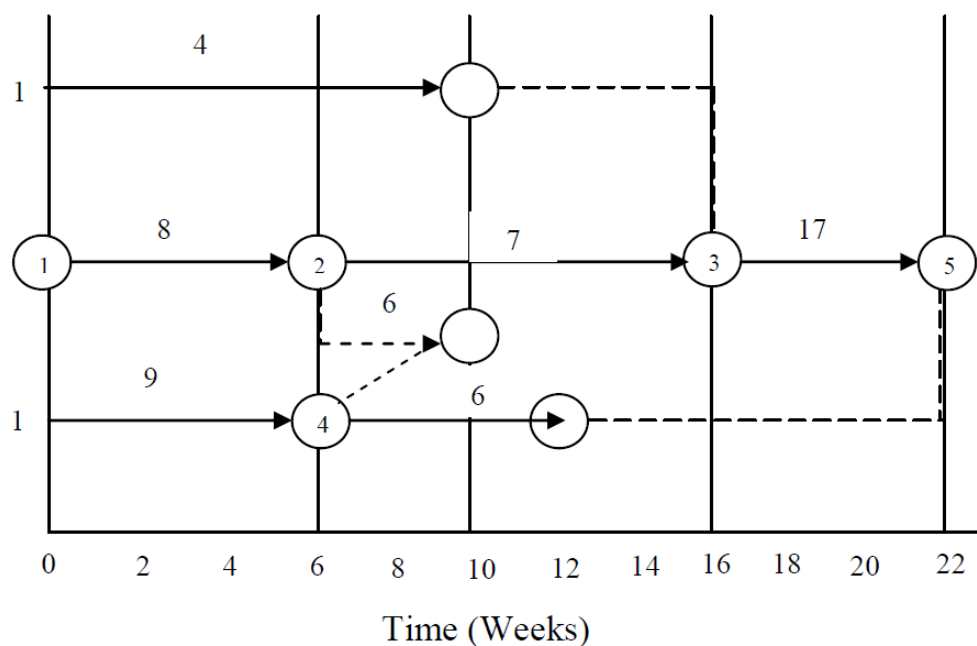
6.2 Resource levelling

A resource may be required to be used in a uniform manner. For instance, in the present-day labour situation one cannot have 100 labourers yesterday, 30 today and 80 tomorrow. Once a labour is hired, it is difficult to hire him. The rate of usage of labour must be uniform. Resource analysis used for this category of problems is called Resource levelling.

Example

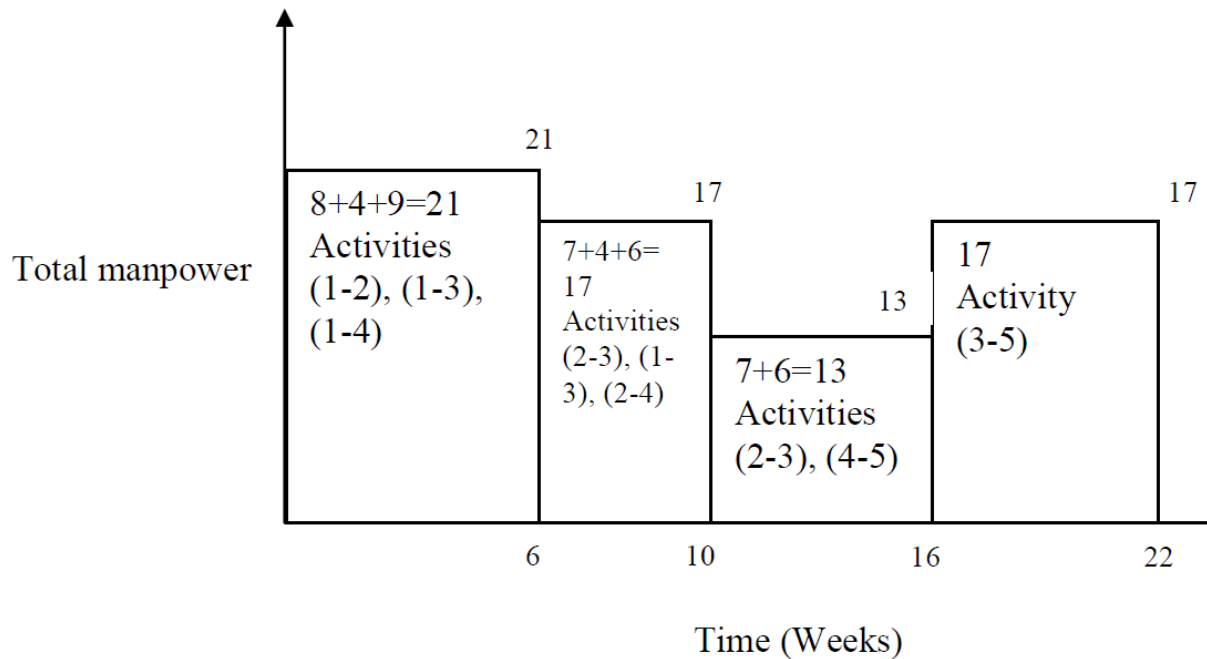
Consider the following problem of project scheduling. Obtain a schedule which will minimize the peak manpower requirement and smooth out period to period variation of manpower requirement.

Activity	Duration in Weeks	Manpower requirements
1-2	6	8
1-3	10	4
1-4	6	9
2-3	10	7
2-4	4	6
3-5	6	17
4-5	6	6

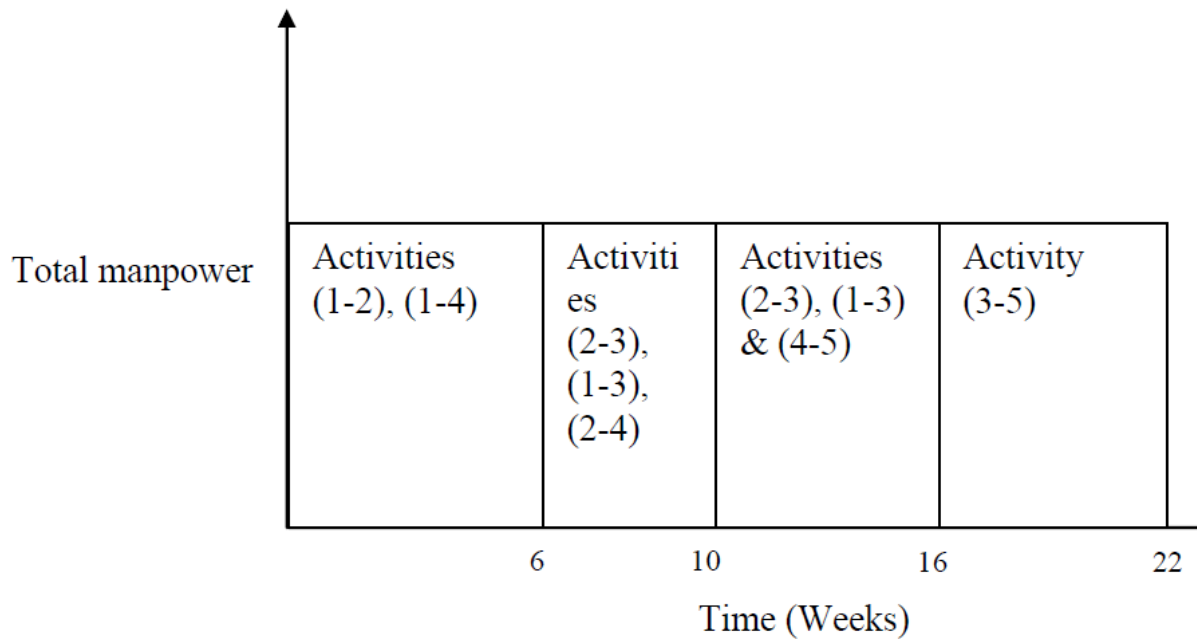




The corresponding manpower requirement histogram is shown below.



From this figure, it is observed that the peak manpower requirement is 21 and it occurs from 0 to 6 weeks. The activities which are scheduled during the period are: (1-2), (1-3) and (1-4). The activity 1-2 is critical activity. So, it should not be disturbed. Between activities (1-3) and (1-4), the activity (1-3) has high slack value of 6 weeks (whereas its only 4 weeks for (1-4)). Hence, it can be started at the end of 6 weeks. The corresponding modification is shown by the following histogram.



Review Questions

- Describe following approaches for 'Make or Buy' decision:
 - Simple Cost Analysis
 - Economic Analysis
 - Break Even Analysis
- Differentiate between EOQ and EPQ inventory models used in Economic Analysis.
- Define forecasting and its basic requirements.
- Compare the quantitative and qualitative forecasting methods.
- Give details of various components of forecasting demand.
- Differentiate between following forecasting methods:
 - Time Series Methods
 - Casual Methods
 - Qualitative Methods
- Explain the forecasting error and accuracy measurement methods.
- Define scheduling and explain its objectives.
- Describe in details operation sequencing and balancing.



10. What are the priority rules to select order of jobs waiting for processing in sequencing?
11. Explain the various steps of using network techniques.
12. Discuss in details the importance of resource allocation and leveling in project planning and control.